LOT COMPUTER ��? BOM

LOT COMPUTER — Components Buying List (BOM)

```
DOCUMENT   LOT-COMPUTER / 02 BOM          REV A   2026-10-11
S-2        VADIK MARMELADOV
LINK COLUMN   VERIFIED = URL returned by a web search this session
              UNVERIFIED = part named, product page not confirmed; search the MPN
PRICES        ESTIMATE only. No live quotes were obtained.
```

PROTOTYPE KIT (buy first — P1, ~2–5 units)

Item	Why	Link	Status
DFRobot DFR1221 — ESP32-S3 1.85” round touch display, Qi receive, mic, 16 MB flash / 8 MB PSRAM	Firmware + API bring-up with Qi built in. Round, not square: firmware only	Core Electronics · DFRobot	VERIFIED
DFRobot FireBeetle 2 ESP32-S3 + OV2640 camera (DFR0975)	Camera bring-up	DigiKey	VERIFIED
BME688 breakout (Adafruit/SparkFun/Pimoroni)	Sensor + AI-gas bring-up	search “BME688 breakout”	UNVERIFIED
SHT45 breakout	Accurate T/RH	search “SHT45 breakout”	UNVERIFIED
Qi charging pad, 5 W	Charger (req. 12)	any WPC-certified 5 W pad	UNVERIFIED

The DFR1221 listing does not mention a camera port; do not expect to attach one.

A. COMPUTE

Part	MPN	Qty	Est. $	Link
Wi-Fi/BLE SoC module, 16 MB flash, 8 MB PSRAM (v1)	ESP32-S3-WROOM-1-N16R8	1	4–6	DigiKey VERIFIED (8-wk lead; one regional page 0 stock)
Bare SoC for 5 mm v2	ESP32-S3FN8 / R8 variants	1	3–5	UNVERIFIED

ESP32-S3 has an 8–16-bit DVP camera interface (confirmed in a listing).

B. DISPLAY & INPUT

Part	Spec	Est. $	Link
Small square colour LCD/OLED ~1.3” with cover glass	fits inside 40×40 mm face, SPI	8–15	UNVERIFIED — choose after D-2
Tactile switch, low-profile (Copy)	≥ 0.5 mm travel	0.3	UNVERIFIED
Status LED (camera-active, hard-wired to camera power)	0402 red	0.05	—

C. SENSORS

Part	MPN	Role	Est. $	Link
Bosch gas+T/RH/P with trainable AI scanner (req. 14, 15)	BME688	weather + air quality	8–10	DigiKey · datasheet (SparkFun mirror, rev 1.0) VERIFIED

Part	MPN	Role	Est. $	Link
Sensirion T/RH	SHT45	precise T/RH, away from heat	2–3	UNVERIFIED
Ambient light	VEML7700	auto-brightness	1–2	UNVERIFIED

Heat note: SoC and charger warm the board; place SHT45/BME688 at an edge with a thermal slot and compensate in firmware.

D. CAMERA

Part	Note	Est. $	Link
OV2640 (2 MP) v1	cheapest; seen on DFR0975	3–6	VERIFIED via DFR0975
OV5640 (5 MP) option	8-bit DVP works on ESP32-S3	5–10	UNVERIFIED
Sub-3 mm micro camera for 5 mm v2	candidates only	TBD	UNVERIFIED

E. POWER & CHARGING

Part	MPN	Note	Est. $	Link
Qi receiver coil	Würth 760308101208 (13 µH, 0.8 A, ø10 mm × 1.7 mm per listing)	Verify match to the Rx IC and that power meets need	1–3	DigiKey VERIFIED
Qi receiver IC	e.g. TI BQ51013B	NOT verified; check coil compatibility	2–3	UNVERIFIED
Li-ion charger + fuel gauge	e.g. BQ25895 / MAX17048	NOT verified	2–4	UNVERIFIED
Thin LiPo 3.7 V, 100–1000 mAh	≤2 mm (v2) / ≤4 mm (v1) with protection circuit	3–6	UNVERIFIED	
Ferrite shield sheet	behind coil	0.5	UNVERIFIED	
Qi charging pad (bundled with each unit)	5 W, WPC	6–10	UNVERIFIED	

F. PCBWAY SERVICES & BODY (req. 1, 3, 17)

Service	Spec	Est. $ per unit @100	Link
PCB fabrication + SMT assembly	4-layer, ~38×38 mm, ENIG	15–30	pcbway.com — quote pages not verified
CNC machining, 316L stainless, 2 parts (back plate, front bezel)	±0.05 mm achievable per third-party guide; walls ≥1.0 mm typical	40–110 pair	PCBWay CNC community examples in 316L — reviews only, not a quote
Mirror polish (back plate)	Ra target ≤0.05 µm	incl./add	**Ask PCBWay**; one reviewer reported poor bead-blast finish
Non-metal Qi/antenna inlay	zirconia ceramic or glass, ~16 mm	5–15	UNVERIFIED
Cover glass	0.5–0.7 mm, AR coat	2–5	UNVERIFIED
Fasteners / adhesive / gasket	M1 screws or laser weld	1–2	—

PCBWay constraint: a steel wall of 1.0 mm is typical minimum; the plan's 0.4–0.6 mm walls are thinner than the cited guidance and must be confirmed with them (another reason 5 mm is a v2 gate).

TOTALS (ESTIMATE)

```
                    LOW      HIGH
Electronics + PCB   $45      $85
Steel body + polish $45      $110
```

```
Charger pad           $6       $10
Glass / inlay / misc  $8       $20
Per unit              ~$120    ~$220     (rounded; excludes labour, shipping, duty)
100 units             $12k     $22k
NRE (prototypes, fixtures, cert)   $15k–$35k
```

PROCUREMENT ORDER

1. Prototype kit this week (long lead times: BME688, ESP32-S3 modules 8–12 wk).
2. PCBWay quote requests (qty 5 and 100) once D-1/D-2 are signed.
3. Samples of the inlay and cover glass during P3.